

Shilpa: Inclusive Mobile Learning App for Visual, Auditory, Physically, and Cognitive Impaired Students

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Abstract— Shilpa is presented as an inclusive mobile learning application designed to support primary education for students with disabilities within a unified platform. Existing educational technologies present limitations, as accessibility features are often implemented in isolation, resulting in fragmented learning experiences. An integrated system was developed to support visual, hearing, physical, and cognitive impairments within a single application environment. Shilpa was developed using mobile-based architecture with integrated accessibility modules, including Braille-based content interaction, sign language recognition, multimodal interaction mechanisms, and adaptive learning support. A Convolutional Neural Network (CNN) model was used for Braille recognition, while a Bidirectional Long Short-Term Memory (BiLSTM) model was implemented for sign language recognition using hand landmark sequences. Shilpa was evaluated through model performance analysis and user-based testing. Experimental results demonstrated high accuracy in Braille recognition and sign language classification, with effective generalization and minimal overfitting. User evaluations demonstrated improved accessibility, interaction efficiency, and learning engagement across different impairment groups. The findings confirm that integrating multiple accessibility features within a single platform enhances usability and supports inclusive learning. Shilpa contributes to reducing accessibility barriers and promoting independent learning in a Sinhala-medium educational context.

Keywords— inclusive education, mobile learning, Braille recognition, adaptive learning, sign language recognition

I. INTRODUCTION

Many learners with disabilities find that digital education tools were simply not built with them in mind. Content formats, interaction methods, and assessment approaches tend to favor the majority, leaving students with visual, hearing, physical, or cognitive impairments to navigate systems that do not work well for them [1], [2]. Mobile learning has grown popular in primary education because of its flexibility and accessibility, but many applications still treat accessibility as an afterthought rather than something built in from the start. This gap persists even in systems that follow established

standards like the Web Content Accessibility Guidelines (WCAG) [3]. When accessibility is instead embedded into the core of a system's design, research has shown that learner engagement genuinely improves. This is the central idea behind the Universal Design for Learning (UDL) framework [4].

Shilpa was developed as an inclusive mobile learning application targeting grades 3 to 5, bringing accessibility support for multiple types of impairments into a single platform. The groundwork for this kind of system had already been laid in earlier research. Sinhala Braille recognition and translation have been shown to be technically feasible [5], [6], and Sinhala Sign Language (SSL) based recognition using computer vision techniques [7]. Interactive and game-based methods were found to improve both sign language learning and learner motivation [8]. Gaze-based control and head-based pointing have been explored to reduce reliance on touch input [9], [10]. Data-driven adaptive learning systems also showed promise in adjusting learning content based on individual learner performance [11].

Most existing systems are developed separately and do not support multiple impairments on a single platform. Shilpa offers Braille content, sign language support for learning, multiple supports for students with motor disabilities, and adaptive learning features. Shilpa supports learning for easier students, guiding in the Sinhala medium. Teachers teach each disabled student various methods.

II. LITERATURE REVIEW

Disability focuses on four main types. Visual impairment, sign-supported learning for hearing-impaired students, multiple interaction methods for physically impaired students, and adaptive instruction for cognitively impaired students. A single system that combines different disabilities is lacking in the educational platform.

Visual impairment focuses on a printed Braille document without losing its structure, making the tactile content readable. Braille conversion treated translation and formatting as core processes, enabling the direct generation of tactile

output from digital text representations [12]. Modular transcription frameworks were developed in which translation and formatting functioned as connected steps using configurable mappings. Braille was also added to Extensible Markup Language (XML) based workflows. It is a simple language that does not require code changes [13]. Classroom use is difficult due to the preparation and checking of materials, which require significant time and effort from teachers [14]. Good Braille accessibility depends on a properly formatted document, not converting individual characters.

Sign language recognition is most beneficial for people with hearing impairments. Datasets achieved strong results, but in the real classrooms, students get different results. Students use a range of signs and gestures that blend into one another, and cameras do not always capture hand movements clearly, so accuracy improves with practice [15]. A large number of signs in a flow could be recognized correctly when the system was trained on more examples and used the method to follow how signs connect and change from one to another [16]. The app is working well and provides accurate recognition; students want to learn the correct ones and need to practice to reduce mistakes in the activities.

Physical impairment support primarily emphasizes maintaining motor function and managing fatigue levels. Gaze-based technologies demonstrate usability in real-world situations when combined with structured evolution and monitoring, thereby supporting long-term accessibility for students with motor impairments [17]. Multimodal interaction systems were introduced to reduce usability limitations for students with physical disabilities. The gaze-voice interaction feature demonstrated effectiveness by using voice as a control method, reducing interaction limitations for maintaining usability [18]. Evidence shows that adaptive selection, multimodal switching, and reduced physical effort are critical to long-term assistive technology.

Cognitive impairment has been studied using various approaches, including Adaptive Learning Systems (ALS), Intelligent Tutoring Approaches, and Engagement-focused Instructional Design. Intelligent Tutoring Systems (ITS), widely used in ALS, have been shown to significantly improve K–12 academic performance when effectively implemented [19]. To optimize K-12 education, diverse systems have been implemented. Gamification approaches have been found to increase engagement when combined with suitable instructional design and evaluation methods [22]. Learning efficiency and flexibility, adaptive microlearning, personalized content delivery, and feedback mechanisms have all been utilized [23]. Artificial Intelligence (AI)- enabled Adaptive Learning (AL) platforms in K-12 education highlight the potential of data-driven personalization to improve learning pathways and outcomes.

III. METHODOLOGY

Shilpa is a unified mobile application learning platform. It supports students with Special Educational Needs and Disabilities (SEND) in grades 3 – 5. It provides support for students with visual, hearing, and physical impairments, as well as students with cognitive learning difficulties. It is a single unified application for use by disabled students. WGAC and UDC shaped the accessibility and instructional design requirements. Functionally, each disabled type provides features suited to that type.

Modules were developed independently before being brought together. No interaction work began until a module had been tested and confirmed in isolation. This kept localization using the Sinhala language. Data collection is underway to create an SSL gesture dataset for hearing-impaired support, and a Braille dataset is being created using a slate and stylus. It is created with the help of blind people. The overall development shows clearly step-by-step paths in Shilpa. It starts to understand what each type of disability wants, then collects data, develops each part, and combines them all. All are combined. After that, check that it works correctly, remains stable for long-term use, and is more effective for students. Fig. 1 shows an overview of the project.

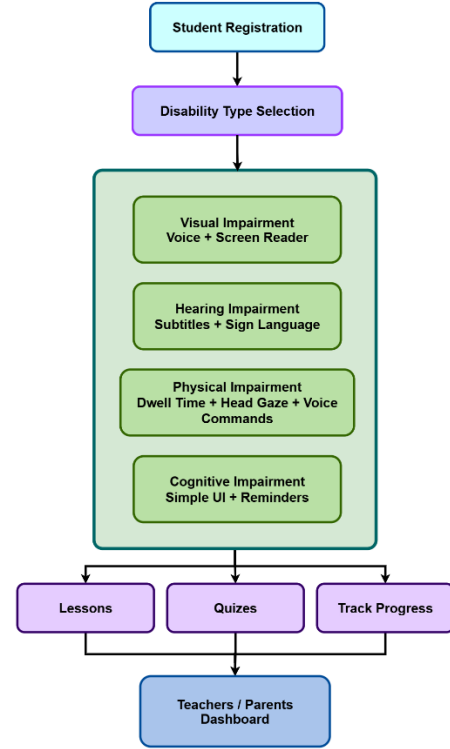


Fig. 1. Overview of the Shilpa inclusive mobile learning system.

A. System Architecture Overview

Shilpa was designed with different features. It is a mobile app. The app was developed using Flutter. The backend was built with Node.js, and MongoDB was used to store data. It uses Representational State Transfer (REST) Application Programming Interfaces (APIs) to connect paths. All four accessibilities share one app in the same environment, and a single unit covers students' needs. Fig. 2 shows the system architecture diagram of the Shilpa app.

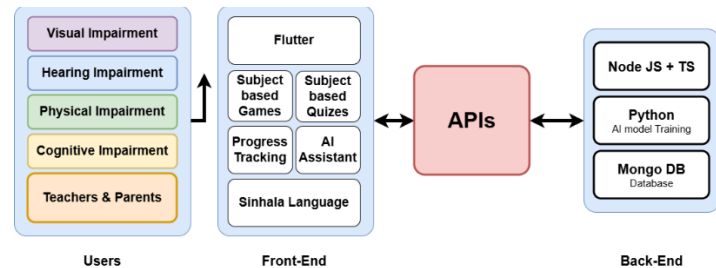


Fig. 2. System Architecture of Shilpa inclusive mobile learning system.

B. Visual Impairment Support

Visual impairment interacts with learning Braille and voice using the teaching concept. The dataset is created manually, and the model for verifying quiz answers is used after students write.

A Convolutional Neural Network (CNN) model was trained for Braille recognition. An image-based dataset for training the model. The CNN model uses layers to find patterns in the Braille dots and reduce the dimensionality of the feature of details.

The feature extraction was converted to a single layer. It is passed to the dense layer for digit identification. A dropout layer is used to reduce overfitting and improve model stability. The final output uses the Softmax function to assign probabilities to the Braille digit classes and returns the class with the highest probability as the predicted digit.

After training was completed, the model was added to the app. The questions PDF given to the student. Parents and teachers can save it as a printable document. Students can answer the questions using a new paper. Parents or teachers can take a photo of the Braille answers and upload it to the system. The CNN model analyzed dot patterns on the answer sheet. The model processes the image and predicts whether the student's answers are correct. Based on the analyzed results, Shilpa evaluated students' performance. Shilpa helps visually impaired students to improve their knowledge.

C. Hearing Impairment Support

The hearing-impaired section helps deaf students learn using sign language. The dataset is created to support the deaf school. The dataset includes SSL word signs, letter signs, and number signs. This was collected to train the recognition model.

Hand landmark was detected using the MediaPipe framework. The dataset gets sign videos to train the model. Movements were given to the Bidirectional Long Short-Term Memory (BiLSTM) model built using TensorFlow.

BiLSTM was good at understanding how signs change and move. It reads the gesture sequence in both forward and backward directions and understands the full meaning of each sign. A dropout layer was added within the network to stabilize training and prevent the model from memorizing the training data. The SoftMax function was used to give a probability score for each SSL gesture.

After training, the model is converted to the TensorFlow Lite format and integrated into the mobile app. The camera captured the student's hand gestures. MediaPipe is used to extract landmark sequences, which are then passed to the training model for prediction.

Lessons were delivered in sign-based videos, quizzes, and games. In sign imitation games, students had to perform the correct sign, and it was a matching activity. Shilpa recorded the gesture, and the student performed and checked it in the trained model. The student got instant feedback on whether the answer was correct. All results and performance details are saved in the database. Teachers and parents can view students' progress.

D. Physical Impairment Support

The physical impairment introduces a multimodal interaction framework designed for students with motor

control difficulties. The system integrates adaptive dwell-time interaction, gaze-based navigation, and voice-based engagement to maintain hands-free control.

The adaptive dwell-time feature dynamically adjusted the engagement time based on user performance. The number of Interaction attempts was monitored, and the dwell duration was updated within default limits to improve selection accuracy.

Eye and gaze control were implemented using facial landmark detection methods. Facial and eye movements were tracked to estimate the user's exact target of interaction. A stability filter was included to reduce noise and ensure that target positions were processed only when the gaze remained stable.

Voice-based navigation allows users to navigate through the application using simple voice commands. Speech recognition converts audio input into text and recognizes the command. Recognized commands were mapped to predefined actions such as navigation, selection, or activity control. A multimodal controller coordinates interaction methods to fix stable system engagement.

E. Cognitive and Learning Difficulties Support

The Cognitive Learning Support (CLS) module evaluates students' performances and delivers personalized learning support. CLS consists of Gamification Assessments (GAs) and the Adaptive Learning (AL) mechanism. The system uses short, interactive GAs designed to meet K-12 objectives, such as shape matching, color identification, and timed object interaction.

During GAs' performance, Behavioral Metrics (BMs) like hint usage, correct responses, task accuracy, and response time are measured and recorded.

These metrics serve as structured input features and are processed by a Machine Learning Model (MLM) deployed on the mobile device via the TensorFlow Lite framework. The MLM predicts the "Learner's Support Level (LSL)," which is then classified into predefined Curriculum Levels (CLs) that represent different support needs.

This model is used for the Shilpa mobile application to evaluate students' CLs. When a student completes a GA, the model categorizes the student based on its prediction.

IV. RESULTS AND DISCUSSION

Shilpa app tested students with different disabilities. Testing covered four disability types. It provides support for visual, hearing, physical, and cognitive impairments.

A. Visual Impairment Support

The visual impairment support module was tested to assess its effectiveness for learners with visual impairment. The module is integrated with a CNN-based Braille recognition model and the Android TalkBack screen reader.

Braille digit images were collected and resized to the same size before training. The dataset is split into training and validation sets. Training accuracy increased slowly and reached 92%. Validation accuracy remained between 88% and 89%. Both accuracy values were close to each other during training. This showed the model learned the data well and did not overfit. Training and validation curves are shown in Fig. 3.

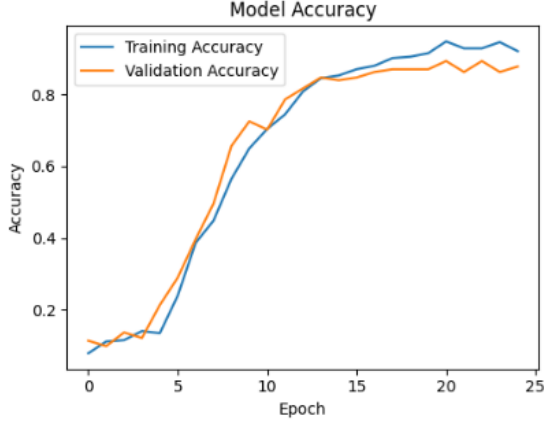


Fig. 3. Training and validation accuracy and loss curves of the Braille number recognition.

This function is useful for both blind and partially blind students. Students can learn using voice-based audio lessons. Students can play voice-based games using swapping options. A Braille question paper was developed to improve students' paper-writing skills. Blind school students were involved in testing Shilpa's visual impairment support function to check if it works well. This helps students to learn and improve their knowledge.

B. Hearing Impairment Support

The hearing module was evaluated through SSL recognition activities, quizzes, and games. During early development, Logistic Regression was applied to sign patterns extracted from hand landmark features. That approach could not capture how signs change over time, so the team moved to a BiLSTM—a sequence model better suited to the temporal nature of SSL gestures.

The training performance of the recognition model is presented in Fig. 4, illustrating training and validation accuracy across epochs. Training accuracy gradually increased, reaching approximately 97–98%, while validation accuracy stabilized at approximately 96–97%. The small difference between training and validation accuracy indicates that the model generalizes well to unseen samples and does not exhibit significant overfitting. The consistent accuracy improvement confirms that the BiLSTM architecture effectively learned temporal relationships in hand landmark sequences required for dynamic sign recognition.

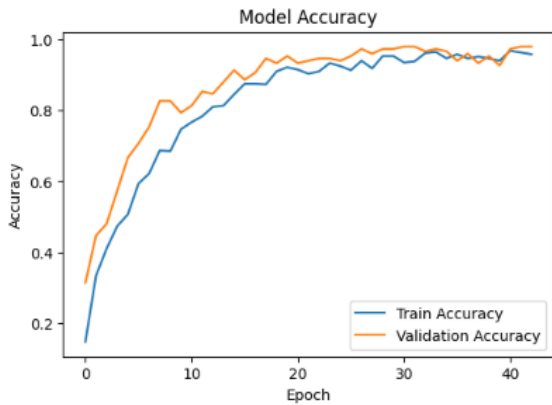


Fig. 4. Training and validation accuracy curves of the BiLSTM SSL model.

Shilpa was evaluated with deaf students during lesson-based activities and sign-based activities quizzes within the mobile application. Students performed sign gestures according to the learning tasks, and Shilpa detected the corresponding signs and provided immediate feedback based on the predicted class. Strong recognition accuracy was maintained when gestures were performed clearly within the camera frame. Recognition errors were observed when hand movements were partially occluded or when gestures were performed rapidly. The evaluation results confirm that the SSL recognition module effectively supported sign-based interaction and enabled accessible learning activities for hearing-impaired learners within the platform.

C. Physical Impairment Support

The physical impairment support module was evaluated to assess the effectiveness of the multimodal interaction framework designed for learners who have difficulty with conventional touch-based mobile interaction. The evaluation was conducted with five students aged 7 – 10 using a smartphone in a classroom environment. Four interaction configurations were tested. Voice-only interactions, gaze-only navigation, gaze with adaptive dwell-time, and a hybrid interaction method combining voice and gaze features. Every section went about 25-30 minutes, and performance was measured by using activity completion time, accuracy of selection, error frequency, and wrong interaction rate.

The adaptive dwell-time feature adjusts the selection duration according to user engagement with the application. As displayed in Table I, the adaptive method reduces task completion time and increases selection accuracy while providing a comfortable interaction between the user and the application. Voice feature achieved good command-recognition accuracy and navigated quickly when commands were clearly captured. Gaze feature interaction provides hands-free control but requires a good stabilization technique to reduce sudden cursor movements.

The results of the interaction modes are presented in Table I. The results show that the hybrid mode, which includes voice and gaze input, achieved the best overall performance score. The hybrid mode reduces interaction issues and false activations compared to individual-modality approaches. The evaluation indicates that the multimodal interaction solution provides more flexible, user-friendly, and usable access for students with motor difficulties.

TABLE I. COMPARISON RESULTS OF INTERACTION MODES

Interaction Mode	Completion Time (s)	Selection Accuracy (%)	Error Rate (%)	False Activation Rate (%)
Voice Only	12.7	91	9	5
Gaze Only	18.4	79	21	16
Gaze Adaptive Dwell	14.2	89	11	7
Hybrid (Voice + Gaze)	10.3	94	6	3

D. Cognitive and Learning Difficulties Support

The CLS model is designed to analyze student learning behavior and provide adaptive learning support. The model

processed BMs from GAs: accuracy, response time, attempts, and hint usage. Utilizing these BMs, students' performance is classified into three Curriculum Levels (CLs): below, average, and above. A synthetic dataset that represents diverse learner behaviors was used to train the classification model. Therefore, to confirm the feasibility, the classification model was validated against pilot datasets. This model achieved 100% classification accuracy on the synthetic test dataset. Moreover, demonstrated its ability to accurately identify different levels of learner performance in controlled scenarios. This verifies that the neural network has effectively learned the relationships between the selected BMs and their corresponding CLs.

On real-world data from 23 students, the model achieved an overall classification accuracy of 82.6%. This indicates the model's strong capacity to assess student performance levels in practical classroom scenarios. Thereafter, a comparison of CLs was performed between the actual and predicted datasets. The model reliably classifies students into below-average and average CLs; it sometimes misclassifies high performers. Classification errors occurred at both average and above-average CLs. Hence, several high-performing students were misclassified as average.

Observed misclassification indicates the need for further studies to accurately distinguish between top students. Identifying this limitation is essential for future model development. These observed misclassification patterns indicate that the performances of the CLs may overlap in real-world learning environments. Variations in learners' attention, response time, and task interaction in practical scenarios are attributed to this misclassification error. Fig. 6 illustrates the accuracy of the model's predictions and the actual student CLs.

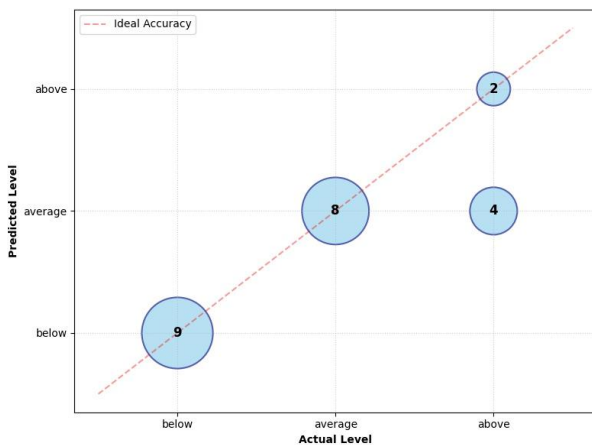


Fig. 5. System Architecture of Shilpa inclusive mobile learning system.

Overall, the classification model identifies learners who need support, empowers educators and stakeholders, and fosters motivation to refine instructional strategies and improve student success.

CONCLUSION

Shilpa was developed as an inclusive mobile learning system designed to support primary school learners with SEND through integrated accessibility modules. Shilpa

incorporated specialized support for visual, hearing, physical, and cognitive impairments, incorporating Braille recognition for visual impairment, SSL recognition for hearing impairment, multimodal interaction mechanisms for physical impairment, and curriculum-level classification for cognitive learning difficulties. Experimental evaluation demonstrated that the proposed approaches enhanced accessibility and supported adaptive learning interactions within Shilpa. The evaluation results confirm that Shilpa enabled learners with diverse needs to engage with educational content more independently. Future work should focus on expanding real-world datasets, conducting larger classroom pilot studies, and further improving the reliability and adaptability of the accessibility modules.

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